

1 Introduction

Each spring, in ecosystems across the temperate zone, ecological communities re-assemble from their period of winter dormancy with the hatching of insects, arrival of migratory birds, the germination of seeds and emergence of new leaves and flowers. The phenology—or seasonal timing—of these re-occurring life cycle events can be quite different, even among closely related organism in a shared habitat (). There is growing evidence that these phenological differences among taxa play a major role in dictating community interactions and shaping the structure and function of ecological communities ().

Phenological differences among species can generate seasonal priority effects; when earlier active species gain a competitive advantage by accessing limiting resources before their competitors emerge or arrive and modify the environment that later emerging species experience (). Studies in plants (), insects (), birds () and mammals () have found that even small phenological priority effects, on the order of just a few days, can substantially alter competitive hierarchies in ways that can persist across several growing seasons ().

The demonstrated importance of (seasonal) priority effects in mediating interspecific competition has led to growing interest in the role of seasonal priority effects in coexistence. Several existing community assembly models that integrate the timing of species arrival or activity have greatly advanced our understanding of the role that temporal dynamics play in coexistence (this was a throwaway place hold because I am not sure exactly what the comment above about timing vs. phenology really means). [It feels natural to mention the competition vs. phenology trade off here, but not dwell on it?]. (Not sure how to connect to climate change here)

But the interest in understanding seasonal priority effects highlights one of the major challenges to the field: phenology is not a static trait. While models focus on the important role of time, the temporal focus on phenology means it must change year to year as it is the realized product of an interaction between the environment (cues) and species unique physiology and genetic and developmental attributes.

Environmental cues are changing rapidly, in fact phenological shifts are one of the most pronounced indicators of climate change to date (). However species' phenological responses to climate change differ significantly. We suggest that the degree to which a species responds to environmental variation known as **phenological sensitivity** (Δ phenology / Δ environmental cue) is the primary mediator of the tradeoff between phenology and competition, and understanding this trait is key to predicting community assembly in an era of global change. Below we outline why integrating phenological sensitivity into community assembly is critical and outline a path to develop a more robust way to predict assembly, especially with environmental change (say better).

2 How phenological differences arise

Environmental cues such as temperature, light and moisture are the primary triggers that control phenology (). Many species respond to multiple aspects of cues. For example some seeds respond to light levels and daylength (). Species are uniquely sensitive to these cues based on physiology, genetics, etc. For example, seeds have different kinds of dormancy. Plants and animals may relay on different cues. What this means is that species individual phenology is response to environment, so the realized phenology differences will vary depending on the environmental conditions.

We illustrate this point by re-analyzing seed phenology germination data from () (Buonaiuto & Wolkovich, see extended Methods for details). 11 co-occurring species were germinated under varying temperature conditions. We can see that even species from a common environment have very different sensitivities to temperature cues (Figure 5a). As seen in our comparison of two species, *Persicaria virginiana* and *Eurybia divaricata*—a pair of woodland herbs that frequently compete in Eastern Temperate Forests ()—we see here that who is first depends on environmental conditions and switches year to year, yet this is not how most coexistence models

approach phenology.

3 Recent work integrating phenology into assembly models

Growing interest in how phenology may structure communities has led to increased modeling efforts, with most approaches centering on the relationship between timing and competition (???). Building off experimental (?) and observational (?) evidence that earlier species appear to perform better models have generally built in trade-off where species with a timing advantage also experience greater competition. Models have done this through an explicit connection of phenological differences to parameters controlling the strength of competition (?), or implicitly by providing a fitness advantage to earlier species that then must be offset in communities where species also vary in parameters related to R^* (??).

These models have all recognized the importance of environmental variability in communities structured in part by phenological differences, generally including interannual variation in the environment that in turn drives differences in phenology. Many models—focused often on grasslands and water competition—have assigned phenology as equivalent to an intrinsic species-level timing (or season length strategy). Year to year variation in the environment then alters absolute differences between species, which then consistently co-varies with resource competition (??). In these approaches the relative orders between species remain fixed, but other models allow species relative orders and timings to vary year-to-year assuming the underlying relationship between pairwise differences in timings and interspecific-competition is known—and static. Each of these approaches in turn thus fixes one aspect of environment $\times$ phenology $\times$ competition relationship, an understandable simplification, but one that may make forecasting community shifts with climate change challenging.

Community assembly models including phenology have all included environmental variability (??) assuming an underlying stationary environment (one that varies year-to-year but with an underlying constant distribution), but climate change represents a clearly non-stationary environment. Temperature, for example, trends higher in its mean over time, with potential additional changes to the variance (?). Because environmental variability in assembly models structures what species combinations are stable, shifts in the environment with anthropogenic climate change are likely to reshape communities, but forecasting them requires accurately capturing how the environment affects phenology and competition together in the simplest—but still robust—form. We argue this means capturing how dynamically the environment can determine both absolute and relative timings of species, and the how those timings directly impact co-existence. While current modeling approaches to date have simplified one aspect of this problem, it can be tractably addressed through new models that explicitly allow these dynamics in both inter- and inter-annual time, and connect to the underlying physiology of phenology.

4 Climate Change, Phenological Sensitivity, Realized Phenology and Co-existence

Predicting phenological responses to a changing environment requires approaches that allow species phenology to vary with the environment based on the underlying physiology. For most species, this means understanding how the environment is shifting and how those shifts filter through species-specific cue systems (Chmura et al 2019 and Bonamour et al. 2019). In addition to warming spring temperatures, phenological phases such as leafout and seed germination that produce seasonal priority effects are influenced by the duration of exposure to that cool winter temperatures (chilling) and integrating this component into models is likely critical for useful assembly forecasts because of its effects on realized phenology.

This variation in timing appears a critical part of most species phenologies and thus likely important to

model, especially given a changing climate ... [maybe a short paragraph here on species all have this sort of variation so may use to inherently bet hedge within years and between years in important ways that other models miss, and then something about the shifting environment and how we need to know that too – the species cue system and how the environment shifts (this could then hint at the future empirical directions would be nice to interweave with the modeling section here if we can.)]

Building on this physiological basis of what determines realized phenology can help us better understand how communities may shift with climate change. Species with high phenological sensitivities will benefit more from adaptive cue shifts but be more compromised by negative cue shifts than less sensitive species, but the full result depends on what stabilizes the community to begin with. If the community is assembled via some aspect of phenology, then a shifting environment could reshape that.

Depending on whether communities encounter and average of 6 or 12 weeks of chilling (Figure 2a), the relationship between phenological sensitivity differences and realized phenological differences shifts (Figure 2c), resulting in different differences (Figure 2b). Because the fundamental tradeoff between realized phenological differences and resource competitive differences does not change in the between environments (Figure 3a), the altered expression of phenology under each environments necessitates you have different communities depending on 6 or 12 weeks (Figure 3a, circled example points). (Now feeling not sure if we need figure 3b).

5 Paths Forward

Predicting assembly in an era of global change will require the closer integration of experiments and theory (?), and enhancing the biological realism of both. Here, we demonstrate a simple way incorporating a more biological realistic germination frame into assembly models that produced important insights about how the relation between phenological sensitivity and competitive ability will shift in order to maintain the relationship between realized phenological difference and competitive differences (Figure Figure 3).

Understanding how shifting environments reshape trade-offs with timing may be especially tractable in grassland systems that have been studied for this topic, but will require new data and approaches to deal with the challenges of cues, which increasing evidence suggests are often interactive. Physiological studies of seed germination highlight this, with multiple temperature cues and soil moisture often affecting germination timing....

Selection for interacting cues is generally thought to be driven by another component likely critical for better predictions—environmental risk. Complex cues are expected to reduce risk [...give an example]. Yet few studies have formally tested this [that I know of]—a major gap that needs to be addressed for modeling. Further, ecological models generally subsume such risks into species traits (e.g., increasing maintenance costs or such), which misses the reality that such risks are a product of timing x environment.

A changing risk landscape may be the underlying driver of observed correlations between germination timing (strategy?) and germination fraction ... [I think there is a lot of interesting bet hedging stuff here that we could close on, and just nod to other life histories there, or better, things beyond plants.]

1. Phenology is not the only thing that varies with the environment trait and density dependent mechanisms (rudoff paper)/ ie, we vary germination fraction and phenology. (note that our model can do this)
2. multiple interacting cues
3. add risk of being early
4. other life histories, i.e, perennials, trees.

Figures

.

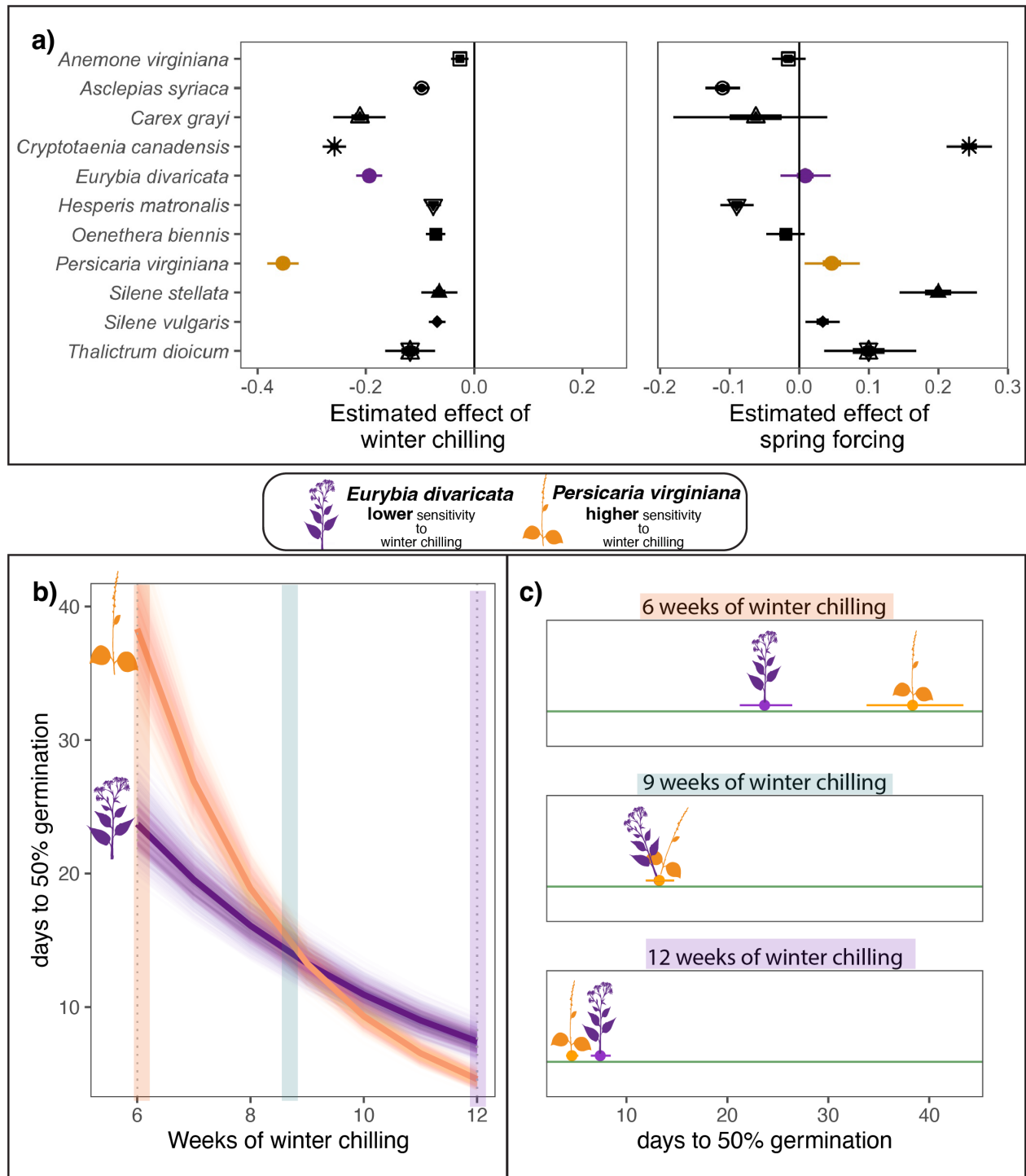


Figure 1: **Co-occur species can differ substantially in their phenological sensitivity (Δ phenological event/ Δ climate) to the environment.** Panel a) estimates the phenological sensitivity of germination timing to winter chilling and spring forcing for eleven herbaceous species of eastern North America. Points represent the mean estimate and thick and thin bars represent the 50 and 95% uncertainty intervals. In b) and c) we highlight the phenological sensitivity to winter chilling of two species *Eurybia divaricata* and *Persicaria virginiana* that are commonly observed in competition. b) Depicts the sensitivity of each species to winter chilling, with thick lines representing the mean estimate and lighter lines representing modeled uncertainty. Panel c) illustrations how the species' different sensitivities manifest in contrasting realized germination differences under different climate conditions. Point represent the predicted mean number of days to 50% germination and bars the 95% uncertainty intervals

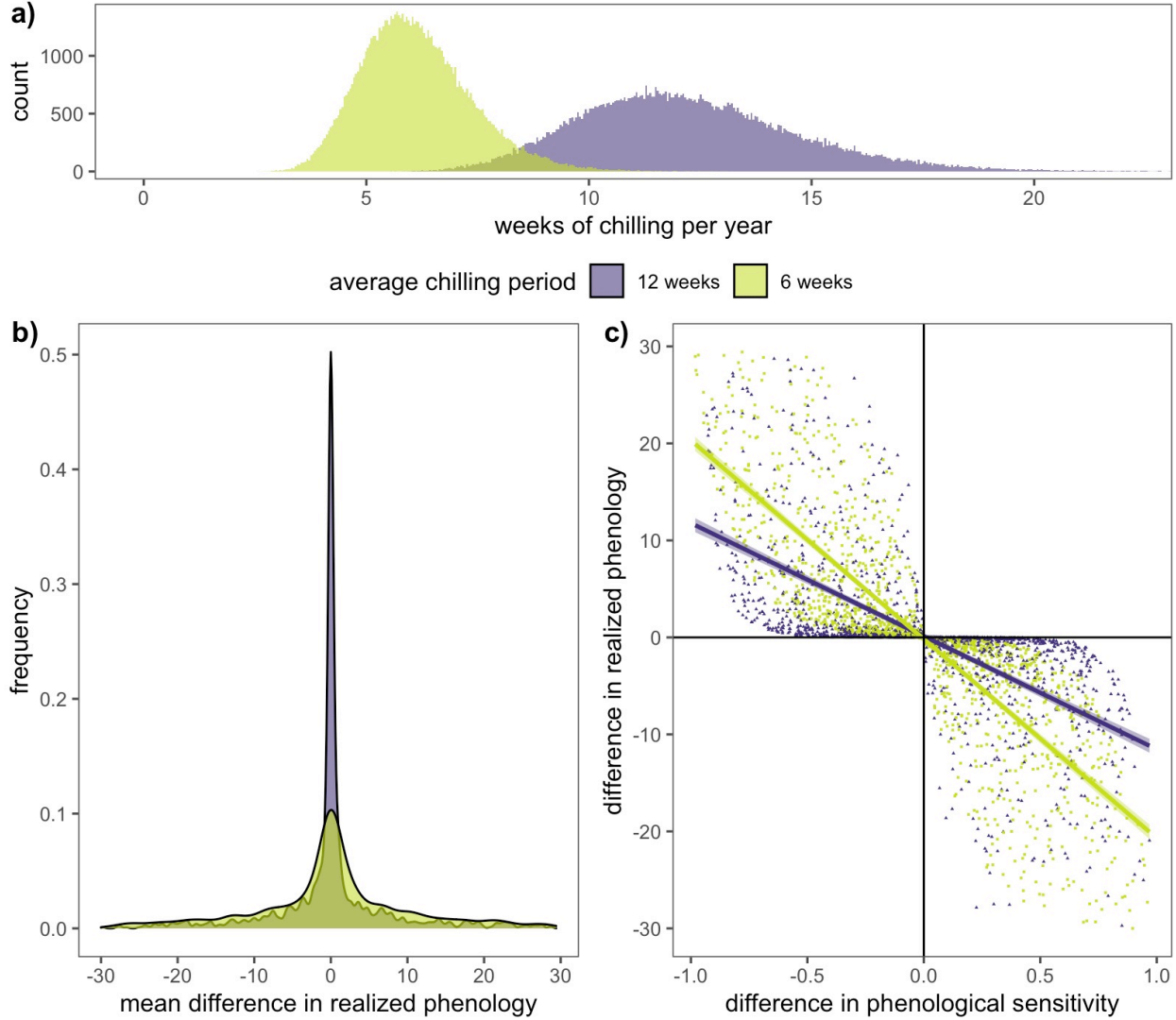


Figure 2: Climate change will the relationship phenological sensitivity differences and realized phenological differences. Panel a) shows how climate change will shift the distribution of the duration of chilling ecological communities receive each winter. Panel b) shows the frequency distributions for the average difference in realized germination phenology under both climate scenarios—shorter chilling durations result in larger realized phenological differences between species. These changes occur because the relationship between phenological sensitivity differences and realized phenology differences, depicted in panel c), are dependent on the environment, and a change in chilling conditions alters this relationship.

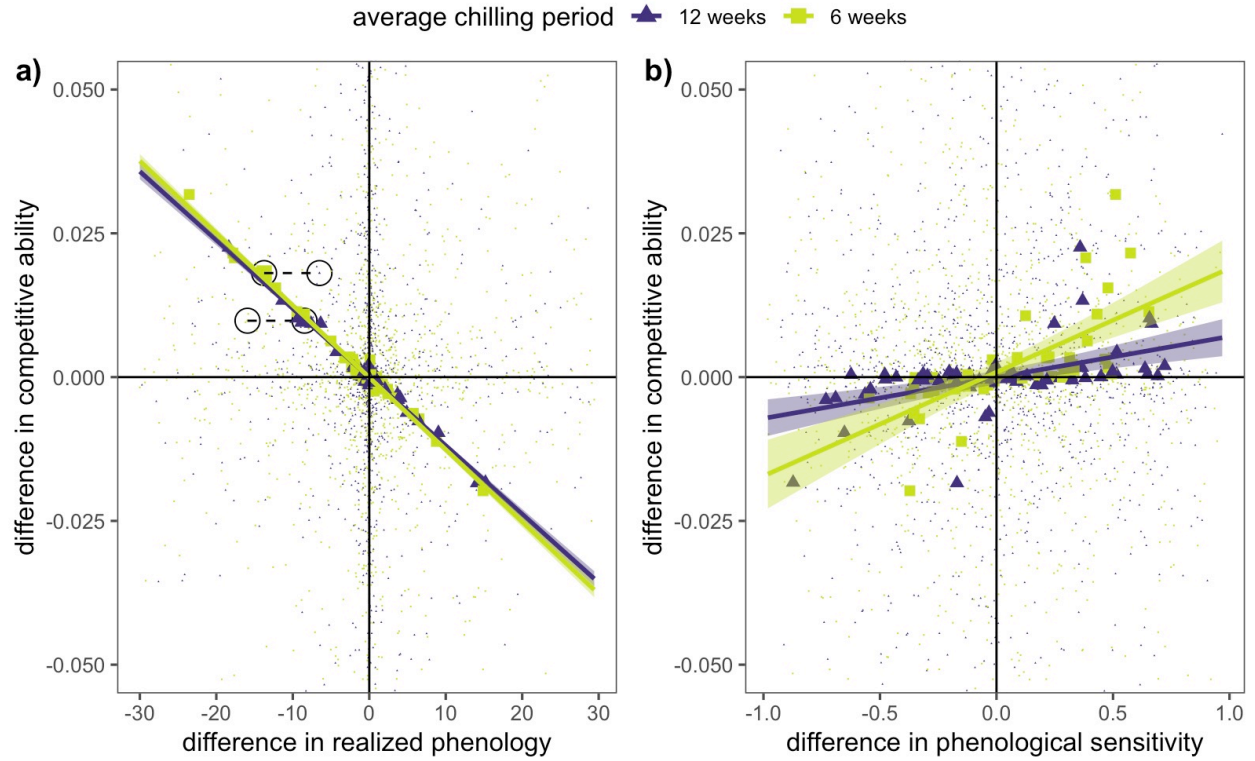


Figure 3: **Climate change will alter the relationship competitive ability, phenological sensitivity, realized phenology and coexistence.** Panel a) depicts the trade-off between realized phenological differences and competitive differences. While the tradeoff relationship between realized phenology and competition does not shift with climate change, species that coexisted under one climate scenario will not coexist if climate conditions shift, as highlighted by the two sets species pairs circled in panel a). Panel b) depicts the trade-off between phenological sensitivity differences and competitive differences under two climate scenarios (average of 12 and 6 weeks of chilling). With climate change, the trade-off between phenological sensitivity and competition does shift because less chilling amplifies the realized phenological differences among species